The Journal of Chemical Physics

RESEARCH ARTICLE | OCTOBER 15 1999

Vibrationally resolved cross sections for single-photon ionization of LiH

Morten F. V. Lundsgaard; H. Rudolph

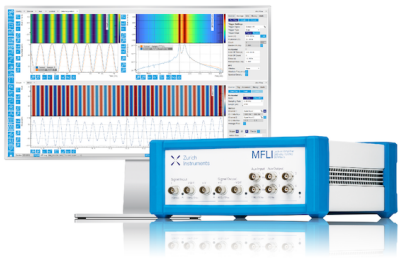
Check for updates

J. Chem. Phys. 111, 6724–6734 (1999)
https://doi.org/10.1063/1.479970

View Online
Export Citation

05 November 2024 20:12:16

Challenge us.
What are your needs for periodic signal detection?
Zurich Instruments
Find out more



AIP Publishing

Vibrationally resolved cross sections for single-photon ionization of LiH

Morten F. V. Lundsgaard and H. Rudolph
Debye Institute, Utrecht University, Princetonplein 5, 3508 TA Utrecht, The Netherlands

(Received 30 March 1999; accepted 27 July 1999)

Ab initio studies of the single-photon ionization of LiH($X\,^1\Sigma^+$) leading to $LiH^+(X\,^2\Sigma^+)$ are reported. The process is studied as function of the internuclear distance (2 a.u.–6 a.u.) and the kinetic energy of the electron (0.05 eV–10 eV). The calculations are based on the so-called iterative Schwinger approach which is implemented within the frozen-core single center approximation. By comparing vibrationally resolved cross sections calculated beyond and within the Franck–Condon principle we find, that the Franck–Condon principle applies relatively well to the present photoionization process. The vibrational wave functions needed for these calculations are obtained from very accurate CI-potentials, also reported in this paper. For LiH the first 10 vibrational spacings are found to agree with the experimental data within 0.2%. Photoelectron spectra (PES) calculated under simulated experimental conditions suggest that photoionization experiments on LiH may yield information on the vibrational structure of LiH^+. The calculated permanent dipole moment of the LiH ground state shows strong influence of the ion-pair channel (Li^+H^-). © *1999 American Institute of Physics.* [S0021-9606(99)30139-2]

I. INTRODUCTION

The LiH molecule is the simplest neutral heteropolar molecule and has as such served as a benchmark system for molecular calculations since the first spectroscopic measurements in the early 1930s.[1,2] The experiments that have followed these pioneering measurements have made it possible to construct reliable Rydberg–Klein–Rees (RKR) (Ref. 3) and inverted perturbation approach (IPA) (Ref. 4) potential curves for the ground state as well as for the first excited states of the neutral molecule. At the same time elaborate *ab initio* calculations for the ground state have been carried out, predicting vibrational level spacings which typically agree within a few percent with the experimental data. The review article of Stwalley and Zemke[5] gives a concise presentation of both the experimental and the theoretical work carried out up to 1993.

Besides the fundamental interest in LiH, its role in the cooling of the premordial gas and its probable influence on the cosmic background radiation[6,7] also makes it of importance in astrophysics. In this context the ionized LiH^+ molecule is also of interest. Compared to LiH the knowledge on LiH^+ is scarce, however. Only in the mass spectrometric experiment of Ihle and Wu,[8] who determined the ionization potential of LiD, has LiD^+ been observed. The rovibrational structure of LiH^+ is thus merely known theoretically. One possible way of getting experimental information would be to measure high resolution kinetic energy spectra of electrons emitted in the photoionization of LiH.

In this paper we study the photoionization of the electronic ground state of LiH,

$$LiH(X\,^1\Sigma^+) + h\nu \rightarrow LiH^+(X\,^2\Sigma^+) + e^-, \tag{1}$$

where the molecular ion is left in its electronic ground state.

This reaction has recently been studied theoretically by Cacelli *et al.*,[9] Berriche and Gadea,[10] and Grabandt *et al.*[11] The latter two calculations were done within the Franck–Condon principle, whereas Cacelli *et al.*[9] reports the photoionization cross section and the asymmetry parameter calculated at the equilibrium internuclear distance of LiH. These studies are extended in the present work by *ab initio* Hartree–Fock calculations of the photoionization cross section and the asymmetry parameter for several internuclear separations (R). From the R-dependent cross sections we have obtained the vibrationally resolved cross sections and, assuming a Gaussian resolution function, photoelectron spectra at temperatures typical for experimental situations.[12] The vibronic states used in these calculations were calculated from accurate *ab initio* CI potential curves for the electronic ground states of the neutral and the ionic molecule, also presented here. The quality of the potential curves is illustrated by the calculated vibrational level spacings for LiH, which typically agree with the experimental observations within 0.2%.

The paper is organized as follows: In Sec. II, the discussion of the calculation of the potential curves and their vibrational structures is followed by a description of the theoretical methods used to calculate the photoionization cross section. The results for the photoionization are presented in Sec. III. Finally, concluding remarks are given in Sec. IV.

II. THEORY

A. Potential curves

The ground state potential curve for LiH($X\,^1\Sigma^+$) and the ground state potential curve for $LiH^+(X\,^2\Sigma^+)$ have both been obtained using the GAMESS-UK program.[13] In Table I the basis sets used in the calculations are listed. The five Li s-functions are contracted from the Li 10 s-basis set,[14] which is augmented with five p- and four d-orbitals. For H the triple zeta ANO basis set is used.[15] Due to the simple electronic structure of LiH it is possible to do a so-called full CI cal-

05 November 2024 20:12:16

 © 1999 American Institute of Physics

TABLE I. Gaussian basis set used for CI calculations of LiH and LiH^+.

Basis function	Center	Type	Coefficient	Exponent
1	Li	*s*	0.00082400	1783.00000000
1	Li	*s*	0.00640300	267.10000000
1	Li	*s*	0.03323900	60.07000000
1	Li	*s*	0.12662100	16.78000000
2	Li	*s*	0.33774900	5.40300000
2	Li	*s*	0.57566900	1.90600000
2	Li	*s*	0.06231100	0.26340000
3	Li	*s*	1.00000000	0.71790000
4	Li	*s*	1.00000000	0.07716000
5	Li	*s*	1.00000000	0.02854000
6	Li	*p*	1.00000000	125.00000000
7	Li	*p*	1.00000000	25.00000000
8	Li	*p*	1.00000000	5.00000000
9	Li	*p*	1.00000000	1.02540000
10	Li	*p*	1.00000000	0.26370000
11	Li	*p*	1.00000000	0.06780000
12	Li	*d*	1.00000000	5.00000000
13	Li	*d*	1.00000000	0.80000000
14	Li	*d*	1.00000000	0.20000000
15	Li	*d*	1.00000000	0.05000000
16	H	*s*	0.00096390	188.61445000
16	H	*s*	0.00749200	28.27659600
16	H	*s*	0.03759540	6.42483000
16	H	*s*	0.14339500	1.81504100
16	H	*s*	0.34863630	0.59106300
16	H	*s*	0.43829740	0.21214900
16	H	*s*	0.16510660	0.07989100
16	H	*s*	0.02102290	0.02796200
17	H	*s*	−0.00131190	188.61445000
17	H	*s*	−0.01034510	28.27659600
17	H	*s*	−0.05049530	6.42483000
17	H	*s*	−0.20738550	1.81504100
17	H	*s*	−0.43508850	0.59106300
17	H	*s*	−0.02472970	0.21214900
17	H	*s*	0.32252600	0.07989100
17	H	*s*	0.70727540	0.02796200
18	H	*s*	0.00242240	188.61445000
18	H	*s*	0.02033820	28.27659600
18	H	*s*	0.08963940	6.42483000
18	H	*s*	0.44229070	1.81504100
18	H	*s*	0.57571440	0.59106300
18	H	*s*	−0.98028900	0.21214900
18	H	*s*	−0.67215380	0.07989100
18	H	*s*	1.14176850	0.02796200
19	H	*s*	−0.01157010	188.61445000
19	H	*s*	−0.08371540	28.27659600
19	H	*s*	−0.44516630	6.42483000
19	H	*s*	−1.14627100	1.81504100
19	H	*s*	2.50318710	0.59106300
19	H	*s*	−1.58284930	0.21214900
19	H	*s*	0.03096569	0.07989100
19	H	*s*	0.30862864	0.02796200
20	H	*p*	0.11279020	2.30500000
20	H	*p*	0.41850750	0.80675000
20	H	*p*	0.47000770	0.28236200
20	H	*p*	0.18262600	0.09882700
21	H	*p*	−0.21086880	2.30500000
21	H	*p*	−0.59437960	0.80675000
21	H	*p*	0.08968890	0.28236200
21	H	*p*	0.86116340	0.09882700
22	H	*p*	0.75995010	2.30500000
22	H	*p*	0.16461590	0.80675000
22	H	*p*	−1.37101400	0.28236200
22	H	*p*	1.05931550	0.09882700
23	H	*d*	0.27051340	1.81900000
23	H	*d*	0.55101250	0.72760000
23	H	*d*	0.33108660	0.29104000
24	H	*d*	−0.79380350	1.81900000
24	H	*d*	−0.09142520	0.72760000
24	H	*d*	0.86200330	0.29104000

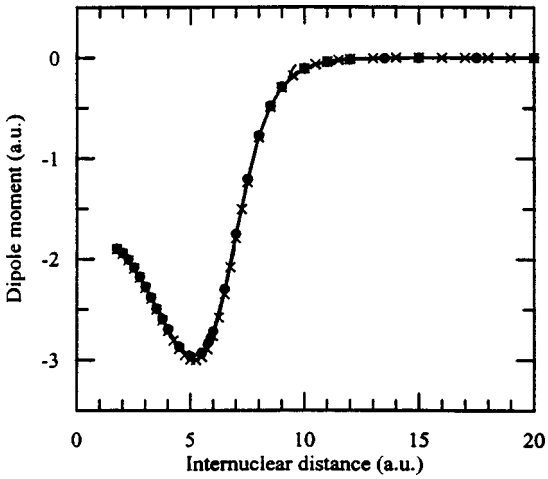


FIG. 1. Permanent dipole moment, μ_X, of LiH($X\,^1\Sigma^+$): crosses, present results; bullets, Partridge and Langhoff (Ref. 20); chain curve, Berriche and Gadea (Ref. 21).

culation. The resulting number of determinants in the two cases are 1 719 306 for the neutral molecule and 52 326 for the ionic molecule. The electron affinity of H and the total electronic energy of Li^+ obtained with the present basis set are 0.026 47 a.u. (0.7203 eV) and −7.276 95 a.u. (−198.016 eV), respectively. These values should be compared to the corresponding experimental values of 0.027 716 1 a.u. (0.754 195 eV) (Ref. 16) and −7.279 834 a.u. (−198.094 47 eV).[17] For the neutral molecule, it was found that the *p*-functions centered on the Li atom are essential for a good description of the potential curve in the region where it is crossed by the Li^+–H^- ionic curve. An estimate for the position of this crossing, $R_x \simeq 6$ a.u., is found by using the zero order approximations $V_{Li^+H^-} = V_{Li^+H^-}(\infty) - 1/R$ and $V_{LiH} = V_{LiH}(\infty)$ for the ionic curve and the ground state curve, respectively. With a slightly more sophisticated model, Mulliken[18] obtained a value of $R_x = 5$ a.u., whereas Boutalib and Gadea[19] in their valence CI calculation report that the diabatic ground state curve and the ionic curve cross at $R_x = 7$ a.u. In Fig. 1, μ_x (calculated with the above mentioned basis set including all excitations, but limiting the valence space to the lowest 50 molecular orbitals, yielding 162 536 determinants) is depicted as function of R (crosses) together with the calculations of Partridge and Langhoff[20] (bullets) and Berriche and Gadea[21] (chain curve). The three sets of data are seen to agree excellently, and the differences are not discernible on the scale of the plot. In all three cases, though, μ_x is seen to vary strongly over the range $R = 2-8$ a.u., due to the strong influence of the ion-pair channel, and it was therefore plausible that this also should affect the photoionization dynamics as a function of internuclear separation. The vibrationally averaged permanent dipole moments are −2.31, −2.36, −2.40, −2.44, −2.48, −2.52, −2.56 a.u. for $\nu = 0-6$, respectively.

Since the highest vibrational states of both the neutral and the ionic molecule are loosely bound, it is important to know the potential curves well at large internuclear distances. Hence, they were calculated for R up to 24 a.u. and 30 a.u. in the LiH($X\,^1\Sigma^+$) and in LiH^+($X\,^2\Sigma^+$) case, re-

05 November 2024 20:12:16

spectively (see Table II). For the calculation of the vibrational states the potential curves must be extrapolated to smaller and larger R values, than those listed in Table II. For $R<1$ a.u. this was done by a combined Morse and Coulomb potential, where the Morse-potential was fitted to the left-hand side of the potential wells and the Coulomb potential, smoothly joined with the Morse-potential, was used in the innermost R-region. For large R, the asymptotic forms of the potentials, $V(\infty)-C_6/R^6$ for the LiH and $V(\infty)-C_4/R^4$ for LiH^+, were matched to the *ab initio* potential curves at the largest R-value. The C-coefficients thus obtained, are $C_6=54.28$ a.u. and $C_4=2.22$ a.u., respectively. The C_4 coefficient is close to the exact value of 9/4 a.u., while the C_6 coefficient differs by 20% from the recommended value of 66.46 a.u.[22] Including higher order terms in the asymptotic expansion of the potentials changes the C-coefficients somewhat, e.g., C_6 becomes 40.2 a.u. when the C_8-term ($C_8=8780$ a.u.) is taken into account. The resulting small changes in the asymptotic behavior of the potentials, does not influence the vibrational energies, however.

The vibrational Schrödinger equation was solved numerically by means of the Numerov method[23] using a step-size of 0.02 a.u. and with a maximum R-value of 40 a.u. and 80 a.u. for the neutral and the ionic molecule, respectively. For the LiH($X\,^1\Sigma^+$) state, the value of the vibrational ground state term was found to be 697.72 cm^{-1} above the equilibrium point in excellent agreement with the experimental value of 697.94 cm^{-1} determined by Chan *et al.*[4] In Table III the term differences, $G_{\nu+1}-G_\nu$, calculated in this work are compared to the theoretical results of Boutalib *et al.*[19] and of Partridge *et al.*,[20] the mixed experimentally/theoretically based data of Stwalley *et al.*,[24] and the experimental data of Chan *et al.*[4] presented in the form of the IPA potential. As the previously reported experimental data[2,3,25–28] are used in the construction of the IPA potential, this has been taken as indicative of the earlier experimental work. The calculations of Boutalib *et al.*[19] were done within a full valence CI model, i.e., the Li atom was treated as an effective one-electron system, whereas Partridge *et al.*[20] included all four electrons in an extensive, but not full, CI calculation. Stwalley *et al.*[24] used a hybrid potential composed of the known long-range form, scaled *ab initio* data of Docken and Hinze,[29] and RKR potentials based on the four lowest vibrational states, to determine the terms.

From Table III the present term differences, $G_{\nu+1}-G_\nu$, are seen to agree with the experimental data[4] within 0.2% for $\nu<10$, which is clearly an improvement of the other theoretical predictions listed in the table. As the vibrational number is increased the agreement between the present results and the experimental data becomes less satisfactory (although still within ~5% for $\nu=20$). The theoretical predictions underestimate the experimental term values, thus indicating that the slope of the potential curve presented in this paper is too small in the asymptotic region. This is consistent with the fact, that the obtained C_6 coefficient is less than the recommended one. The better agreement (~1%) between the results of the valence CI calculation reported by Boutalib *et al.*[19] and the experimental results for the highly excited vibrational states suggests, that the valence CI calculation describes the asymptotic behavior of the potential curve better than the present calculation.

In accordance with the RKR calculations of Stwalley *et al.*[24] we find that the $X\,^1\Sigma^+$ electronic state supports 24 bound states. We note, though, that the calculated $\nu=23$ state is merely bound with 1 cm^{-1}, and it is extremely sensitive to any changes in the shape of the potential curve and in the dissociation energy.

For completeness the spectroscopic constants for the $X\,^1\Sigma^+$ state of LiH are listed together with the experimental data and the theoretical results in Table IV.[3,19,20,24,30] Consistent with the method of Orth *et al.*,[3] we have obtained the vibrational constants from the present vibrational level spacings and from the vibrational level spacings of Boutalib *et al.*[19] and Partridge *et al.*[20] by fitting the vibrational level spacings for $\nu=0-11$ to a fourth order polynomial in $(\nu+1/2)$.[19,20] α_e was calculated from the difference between the rotational constants B_ν and B_e, where B_ν is given by the expectation value of $1/R^2$. Except for the small $\omega_e y_e$ and $\omega_e z_e$ coefficients the agreement is within 2%–3% between the experimental and the present theoretical results. This is also true for the dissociation energy D_e, where the recommended experimental value is 2.515 eV.[27]

For the LiH^+ ionic molecule there are no experimental data for the vibrational structure available, and hence only theoretical results can be compared. In Table V the present results for the vibrational level spacings are compared with the CI calculations of Grabandt *et al.*[11] and Dalgarno *et al.*[6] and with the one-electron model calculations of Berriche and Gadea.[10] The consistency of the various predictions is markedly poorer than in the case of the neutral molecule, e.g., does Grabandt *et al.*[11] find four bound states, Dalgarno *et al.*[6] find six, whereas Berriche *et al.*[10] and we find seven. In their predictions of ΔG_{ν^+} the latter three calculations differ with up to 20%.

In a general paper on the spectroscopic constants of ionic diatomic hydrides Rosmus and Meyer,[31] using the coupled electron pair approach (CEPA), report the following values for LiH^+: $R_e=4.14$ a.u., $\alpha_e=0.484$, $\omega_e=443.4\ cm^{-1}$, and $\omega_e x_e=42.1\ cm^{-1}$. The spectroscopic constants calculated in the present work, $R_e=4.136$ a.u., $\alpha_e=0.507$, $\omega_e=442.9\ cm^{-1}$, and $\omega_e x_e=42.3\ cm^{-1}$, compare well with these data.

B. Photoionization

Within the dipole approximation the transition amplitude, $I_{\mathbf{k},\hat{\mathbf{n}}}$, for the photoionization reaction (1) may be written as (length form)

$$I_{\mathbf{k},\hat{\mathbf{n}}}=k^{1/2}\langle\Psi_i|\hat{\mathbf{n}}\cdot\mathbf{r}|\Psi_{f,\mathbf{k}}^{(-)}\rangle, \tag{2}$$

where Ψ_i is the electronic wave function of LiH($X\,^1\Sigma^+$) and $\Psi_{f,\mathbf{k}}^{(-)}$ is the scattering wave function of the $LiH^+(X\,^2\Sigma^+)+e^-$ system. $\mathbf{r}$ denotes the electronic position operator, $\mathbf{k}$ denotes the momentum of the emitted electron, and $\hat{\mathbf{n}}$ is the direction of polarization of the light. The bound state wave function, Ψ_i, used in the evaluation of the transition moment Eq. (2) was calculated at the Hartree–Fock level using the GAMESS-US program[32] with the basis set listed in Table VI,

05 November 2024 20:12:16

TABLE II. Ground state potential curves for LiH and LiH^+.

R (a.u.)	V_{LiH} (a.u.)	V_{LiH^+} (a.u.)
1.00	−7.376935317	−7.112998437
1.25	−7.655883598	−7.370493869
1.50	−7.827163560	−7.525267948
1.75	−7.932605069	−7.621837049
2.00	−7.996387986	−7.682907211
2.25	−8.033647229	−7.721654820
2.50	−8.053998684	−7.746156344
2.75	−8.063575154	−7.761486136
3.00	−8.066308039	−7.770884366
3.25	−8.064735713	−7.776450227
3.50	−8.060509443	−7.779557556
3.75	−8.054712006	−7.781110914
4.00	−8.048058236	−7.781706318
4.25	−8.041021847	−7.781734076
4.50	−8.033917819	−7.781446692
4.75	−8.026956780	−7.781004530
5.00	−8.020280784	−7.780506575
5.25	−8.013987391	−7.780011199
5.50	−8.008145008	−7.779550244
5.75	−8.002802837	−7.779138631
6.00	−7.997995947	−7.778780874
6.25	−7.993746699	−7.778475441
6.50	−7.990063065	−7.778217639
6.75	−7.986936042	−7.778001453
7.00	−7.984337671	−7.777820673
7.25	−7.982222191	−7.777669503
7.50	−7.980530785	−7.777542884
7.75	−7.979198363	−7.777436495
8.00	−7.978160649	−7.777346788
8.25	−7.977359019	−7.777270808
8.50	−7.976743120	−7.777206224
8.75	−7.976271501	−7.777151087
9.00	−7.975911012	−7.777103834
9.25	−7.975635661	−7.777063145
9.50	−7.975425329	−7.777027972
9.75	−7.975264566	−7.776997453
10.00	−7.975141572	−7.776970864
10.25	−7.975047362	−7.776947606
10.50	−7.974975064	−7.776927195
10.75	−7.974919509	−7.776909215
11.00	−7.974876739	−7.776893327
11.25	−7.974843744	
11.50	−7.974818233	
12.00	−7.974783092	−7.776545575
13.00	−7.974748563	−7.776814770
14.00	−7.974734927	−7.776794110
15.00	−7.974728997	−7.776779804
16.00	−7.974725990	−7.776769628
17.00	−7.974724390	−7.776762222
18.00	−7.974723340	−7.776756722
19.00		−7.776752569
20.00	−7.974722278	−7.776749385
21.00	−7.974721995	−7.776746910
22.00	−7.974721800	−7.776744963
23.00		−7.776743413
24.00	−7.974721575	−7.776742166
25.00		−7.776741155
26.00		−7.776740328
27.00		−7.776739646
28.00		−7.776739079
29.00		−7.776738605
30.00		−7.776738206
⋮	⋮	⋮
∞	−7.974721285	−7.776735464

giving an equilibrium position of $R=3.035$ a.u. and $E=-7.986\,56$ a.u. As shown by Lucchese *et al.*[33] in their study of photoionization of N_2, the use of CI wave functions leads to a better agreement between calculations based on the length (L) and the velocity (V) form of the dipole moment, than is obtained by using HF wave functions. However, by forming the geometric mean of the photoionization cross section from the results obtained with the L-form and the V-form of the dipole moment, the lack of correlations in the HF wave function is partly eliminated (see Hansen[34]). Further, at $R=6$ a.u. the CI-coefficient for the ground state Hartee–Fock reference configuration is 0.92, indicating that the use of the HF wave function in the calculation of the transition moment is reasonable.

For the scattering wave function the so-called *frozen core* approximation was adopted, i.e., the core of the molecular ion is assumed to be identical to the $(N-1)$-electron core of the initial neutral molecule and the scattering wave function (spatial part) may be written as $\Psi_i=\Psi_{N-1}\phi_{f,\mathbf{k}}^{(-)}$, where Ψ_{N-1} is the wave function for the $(N-1)$-electron frozen core and $\phi_{f,\mathbf{k}}^{(-)}$ is the one-electron wave function representing the emitted electron. Within this model $\phi_{f,\mathbf{k}}^{(-)}$ is the regular solution to the Schrödinger equation,

$$\{-\tfrac{1}{2}\nabla^2-1/r-\tfrac{1}{2}k^2\}\phi_{f,\mathbf{k}}^{(-)}(\mathbf{r},R)=-V_{N-1}(\mathbf{r},R)\,\phi_{f,\mathbf{k}}^{(-1)}(\mathbf{r},R), \tag{3}$$

where $\mathbf{r}$ is the position of the photoelectron. The term involving the short range part, $V_{N-1}(\mathbf{r})$, of the static-exchange potential due to the $(N-1)$-electron core is here considered as an inhomogeneous term to the Schrödinger equation for the pure Coulomb scattering problem.

Equation (3) was solved using the iterative Schwinger variational method, which in recent years has been successfully applied to both electron–molecule collisions and to molecular photoionization.[35–37,33] A detailed description of the method is given by Lucchese *et al.*[38] Here a single-center expansion version of the iterative Schwinger variational approach was used. This requires that all functions, operators, and matrix elements are expanded in terms of spherical harmonics, all centered at the same origin, here taken as the center of mass (CM) of the nuclei. A maximum of 40 partial waves was used in the expansion of the bound molecular orbitals, the direct potential and the exchange potential. For the Coulomb interaction, $1/r_{ij}$, 80 and 40 partial waves were used for the direct and the exchange term, respectively. All matrix elements were calculated on a radial grid with 340 points extending out to 70 a.u. with a minimum grid distance of 0.02 a.u. The single center expansion of the bound state is described in Table VII.

The scattering wave function was expanded in nine partial waves,

$$\phi_{f,\mathbf{k}}^{(-)}(\mathbf{r},R)=(2/\pi)^{1/2}\sum_{lm} i^l\psi_{f,klm}^{(-)}(\mathbf{r},R)Y_{lm}^*(\hat{\mathbf{k}}), \tag{4}$$

where the l-summation runs from 0 to 8. For each partial wave the Lippmann–Schwinger form of Eq. (3) is solved iteratively by first expanding the photoelectron wave function, $\psi_{f,klm}^{(-)}(\mathbf{r},R)$, within a given set of basis functions $\{\boldsymbol{\alpha}\}$. The partial wave solutions of Eq. (3) thus obtained are then

05 November 2024 20:12:16

TABLE III. Spacing, ΔG_{ν^+}, between the vibrational levels of LiH.

ν	Present[a]	Boutalib *et al.*[a]	Partridge *et al.*[a]	Stwalley *et al.*[b]	Chan *et al.*[c]
0	1359.66	1344.77	1352.9	1356.6	1359.71
1	1314.68	1300.34	1309.0	1306.2	1314.89
2	1270.55	1256.70	1266.5	1274.3	1270.89
3	1227.31	1214.96	1223.6	1231.4	1227.77
4	1184.87	1174.65	1180.1	1194.6	1185.44
5	1143.06	1135.03	1135.6	1156.7	1143.77
6	1101.72	1095.43	1091.4	1115.5	1102.60
7	1060.73	1055.45	1050.9	1074.4	1061.78
8	1019.88	1015.19	1009.6	1032.9	1021.17
9	978.85	974.80	969.9	990.4	980.52
10	937.40	934.18	929.4	947.4	939.62
11	895.21	893.04	888.3	903.0	898.09
12	851.75	850.97	845.5	857.9	855.48
13	806.39	807.46	800.0	811.4	811.19
14	758.32	761.63	751.8	762.9	764.44
15	706.47	712.37	700.0	710.7	714.22
16	649.46	658.25	643.1	654.0	659.27
17	585.50	597.54	578.5	591.4	597.89
18	512.30	528.01	503.1	520.7	527.91
19	427.12	446.88	415.2	437.3	446.61
20	326.95	350.41	312.0	337.5	350.99
21	209.30	235.72	189.5	221.2	237.71
22	76.29			78.0	

[a]Theoretical results.
[b]Results partly based on experimental data.
[c]Results obtained from experimental data.

05 November 2024 20:12:16

added to $\{\boldsymbol{\alpha}\}$, and Eq. (3) is solved with this enlarged basis set, which then is replaced by $\{\boldsymbol{\alpha}\}$ and the new set of solutions, and so forth. The basis functions $\{\boldsymbol{\alpha}\}$ used in this work are listed in Table VIII. After two iterations the convergence for the total cross section is estimated to be better than 2% at all internuclear distances and electron energies. The scattering wave function was calculated for 17 electron energies ranging from 0.05 eV to 10 eV, and for 11 internuclear separations varying from 2 a.u. to 6 a.u. As the equilibrium position of the ground state of LiH is ~3 a.u. (see Table IV), this R-range covers the classical regions of all but the most highly excited vibrational states.

In the partial wave expansion, the transition moment Eq. (2) is given by

$$I_{\mathbf{k},\hat{\mathbf{n}}} = (4\pi/3)^{1/2} \sum_{lm\mu} I_{lm\mu}(k,R) Y^*_{lm}(\hat{\mathbf{k}}) Y^*_{1\mu}(\hat{\mathbf{n}}), \tag{5}$$

where

$$I_{lm\mu}(k,R) = k^{1/2} \langle \Psi_i | r Y_{1\mu}(\hat{\mathbf{r}}) | \psi^{(-)}_{f,klm} \rangle. \tag{6}$$

$|I_{\mathbf{k},\hat{\mathbf{n}}}|^2$ gives, except for a constant, the cross section for photo-ionization at a given internuclear distance as function of the direction of the polarization of the photon with respect to the internuclear axis, and as function of the direction and the energy of the emitted electron. By averaging over both molecular orientation and emission direction of the electron, we obtain the total photoionization cross section.

TABLE IV. Spectroscopical constants for LiH ($X\ ^1\Sigma^+$). All constants are given in cm^{-1}, except for D_e (eV) and R_e (a.u.).

ω_e	$\omega_e x_e$	$\omega_e y_e$	$\omega_e z_e$	B_e	α_e	D_e	R_e
1406.18[a]	23.5777	0.23948	−0.0069	7.516	0.2124	2.4924	3.0148
1405.86[b]	23.3718	0.21496	−0.0056	7.5161	0.21894		3.0149
1401.5[c]	22.5			7.48	0.212	2.48	3.022
1391.29[d]	23.8597	0.38073	−0.0133			2.5013	3.007
1394.79[e]	20.8563	−0.15565	0.0095			2.4762	3.026
1409.73[f]	25.5818	0.71508	−0.0302				

[a]Present results.
[b]Orth and Stwalley (Ref. 3).
[c]Meyer and Rosmus (Ref. 30).
[d]Boutalib and Gadea (Ref. 19).
[e]Partridge and Langhoff (Ref. 20). R_e is obtained from the potential curve listed in Table I of Partridge and Langhoff (Ref. 20).
[f]Stwalley *et al.* (Ref. 24).

TABLE V. Spacing, ΔG_{ν^+} in cm^{-1}, between the vibrational levels of LiH^+.

ν^+	Present	Berriche *et al.*[a]	Grabandt *et al.*[b]	Dalgarno *et al.*[c]
0	351.6	357.4	335	354.1
1	257.2	265.6	245	261.6
2	163.5	173.0	152	169.5
3	84.1	92.2	⋯	90.3
4	31.8	37.5	⋯	35.6
5	7.3	9.2	⋯	⋯

[a]Reference 21.
[b]Reference 11.
[c]Reference 6.

$$\sigma(k,R)=\frac{4\pi^2 E_{\rm ph}}{3c}\sum_{lm\mu}|I_{lm\mu}(k,R)|^2. \tag{7}$$

The photon energy $E_{\rm ph}$ is related to the electron energy $k^2/2$ by $E_{\rm ph}=\Delta\varepsilon+k^2/2$, where $\Delta\varepsilon=\varepsilon_{\nu^+}-\varepsilon_{\nu}$ is the energy difference between the final and the initial state.

By only averaging over all molecular orientations with respect to the polarization axis, the so-called integrated target angular distribution (ITAD) cross section[38] is obtained

$$\frac{d\sigma}{d\Omega}(k,R)=\frac{\sigma(k,R)}{4\pi}[1+\beta(k,R)P_2(\cos\theta)]. \tag{8}$$

Here θ is the angle of the ejected electron with respect to the direction of the polarization of the incoming linearly polarized photon. The asymmetry parameter β can be expressed in terms of the partial wave transition moments $I_{lm\mu}$, see, e.g., Lucchese *et al.*[38]

So far the rovibrational part of both the initial and the final molecular wave function has been ignored. Since the present experimentally obtainable energy resolution in photoelectron spectroscopy allows specific transitions between rovibrational states to be identified, it is of interest to calculate the rovibrationally resolved cross sections. Although some angular momentum transfer will take place between the photoelectron and the $LiH^+(^2\Sigma^+)$ ion state we assume that only transitions between rovibrational states $|\nu N\rangle$ with the same rotational quantum number N take place. Denoting the rovibrationally resolved cross sections by $\sigma_{\nu N,\nu^+N^+}$, we have with $N^+=N$,

$$\sigma_{\nu N,\nu^+N}(E_{\rm ph})=\frac{4\pi^2 E_{\rm ph}}{3c}\sum_{lm\mu}|\langle\nu N|I_{lm\mu}(k,R)|\nu^+N\rangle|^2. \tag{9}$$

TABLE VI. Gaussian basis set employed for the SCF calculation of the $Li(X\,^1\Sigma^+)$, that is used in the determination of the photoionization cross section.

Center	Type	Exponent
Li	*s*	1783.0, 267.10, 60.070, 16.780, 5.403
Li	*s*	1.906, 0.2634, 0.7179, 0.07716, 0.02854
Li	*p*	1.0254, 0.2637, 0.0678
Li	*d*	0.8000, 0.2000, 0.0500
H	*s*	33.640, 5.0580, 1.1470, 0.3211, 0.1013
H	*p*	4.0000, 1.0000, 0.2500

TABLE VII. Single center expansion coefficients (in percentage) for the occupied orbitals of $LiH(X\,^1\Sigma^+)$ as function of the internuclear separation *R*.

R (a.u.)	Orbital	$l=0$	$l=1$	$l=2$	$l=3$	$l=4$	Sum
2.0	1σ	87.6	11.2	1.0	0.1	0.0	99.9
	2σ	47.9	45.3	4.5	1.5	0.5	99.7
4.0	1σ	64.5	27.5	6.1	1.3	0.4	99.8
	2σ	45.2	33.1	11.3	5.4	2.5	97.5
6.0	1σ	45.1	35.1	13.0	4.3	1.5	99.0
	2σ	39.1	22.7	15.2	9.8	5.6	92.4

In the following we often consider states with $N=0$ and in order to keep the notation simple, we therefore write $\sigma_{\nu\nu^+}=\sigma_{\nu 0,\nu^+ 0}$.

III. RESULTS

The cross section and the asymmetry parameter for photoionization of LiH have previously been calculated by Cacelli *et al.*[9] at the experimentally determined equilibrium distance of 3.015 a.u. The cross section and the asymmetry parameter for photoionization of LiH have been calculated using both the length (L) and velocity (V) form of the dipole operator. In Fig. 2(a) the results for the cross section, σ, obtained at the equilibrium distance 3.015 a.u. using the length form (long dashed curve) and the velocity form (dashed curve) are shown. As pointed out by Hansen,[34] the geometric mean, $\sigma^G=\sqrt{\sigma^L\sigma^V}$, of the cross sections is only sensitive in second order to the lack of correlation, hence it is also depicted in Fig. 2(a) (full curve). $\Delta\varepsilon$ has been determined using the recommended ionization potential I.P. of 7.7 eV.[39] Except near threshold the dependence on the photon energy of σ^L and σ^V is practically identical, with σ^L being constantly ~40% larger than σ^V. The random phase approximation (RPA) results of Cacelli *et al.*[9] for σ^L and σ^V at $R=3.015$ a.u., shown in Fig. 2(a) as double and single-chain curves, respectively, are seen to fall in between the present SEA results, almost on top of the σ^G curve. Cacelli *et al.*[9] found a similar picture in their comparison of RPA- and

TABLE VIII. Gaussian basis set used in the iterative Schwinger method for determining the continuum wave function.

Channel	Center	*l*	*m*	Exponents
σ	Li	0	0	8.0, 4.0, 2.0, 1.0, 0.5, 0.25
	Li	1	0	1.0, 0.5, 0.25
	H	0	0	8.0, 4.0, 2.0, 1.0, 0.5, 0.25
	H	1	0	1.0, 0.5, 0.25
	CM	1	0	1.0
	CM	2	0	1.0
	CM	3	0	1.0
	CM	4	0	1.0
π	Li	1	1	8.0, 4.0, 2.0, 1.0, 0.5, 0.25
	Li	2	1	1.0, 0.5, 0.25
	H	1	1	8.0, 4.0, 2.0, 1.0, 0.5, 0.25
	H	2	1	1.0, 0.5, 0.25
	CM	1	1	1.0
	CM	2	1	1.0
	CM	3	1	1.0
	CM	4	1	1.0

05 November 2024 20:12:16

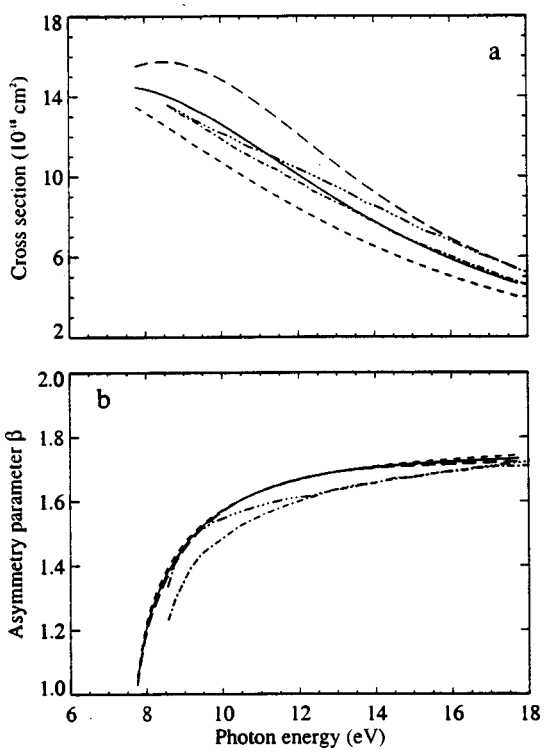


FIG. 2. (a) Photoionization cross section and (b) asymmetry parameter, β, as function of photon energy. R is kept fixed at the equilibrium distance of LiH. Present calculations: dashed curve, velocity form; long dashed curve, length form; full curve, geometric mean of velocity and length form. RPA calculations from Cacelli *et al.* (Ref. 9): broken curve, velocity form; double broken curve, length form.

SEA-calculations for photoionization of Li_2, but unfortunately they did not report any SEA results in the LiH case. For the asymmetry parameter [Fig. 2(b)], however, the present SEA results are less sensitive to the choice of form for the dipole operator, than the RPA results of Cacelli *et al.*[9] In general, though, the agreement between the two types of calculations is reasonable.

In Fig. 3 the absolute value of the dipole matrix, $r^V_{lm\mu}$, element is shown as function of the electron energy for various partial waves of the scattering wave function at $R=2.5$ a.u. and $R=4.5$ a.u. From Fig. 3, we note that for all electron energies and internuclear separations the π-channel is dominant. At small internuclear distances, the character of the scattered wave mainly has d-character, which is consistent with the large p-fraction of the 2σ-orbital of the initial LiH($X\,^1\Sigma^+$) state, see Table VII, based on a propensity for $\Delta l=+1$ transitions. As the distance between the nuclei is increased, higher order partial waves of the 2σ-orbital appear, and the photoelectron wave function exhibits a significant f- and g-character.

The R and $E_{\rm ph}$ dependent cross section, see Eq. (7), is shown in Fig. 4. For energies just above the threshold of 7.7 eV, $\sigma^G(E_{\rm ph},R)$ increases smoothly from ~10 Mb at $R=2$ a.u. to a peak value of ~20 Mb about $R=4.5$ a.u. We note that this is very similar to the behavior of the absolute value of the permanent dipole moment, $|\mu_X|$, of the initial state as R is varied. Hence, it appears as if the photoionizing process close to the ionization threshold is strongly influenced by the ionic nature of the neutral molecule. With increasing photon energy the variation of σ^G with R becomes less pronounced.

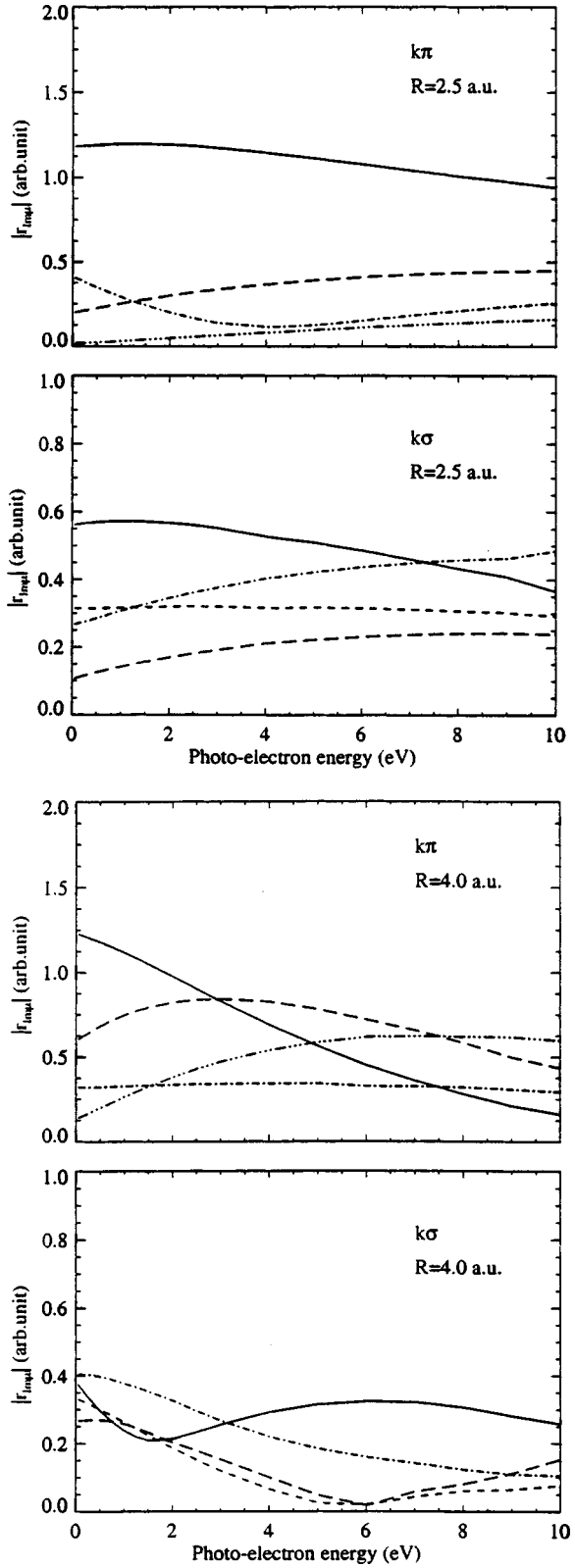


FIG. 3. Dipole matrix element (velocity form), $|r^V_{lm\mu}|$, as function of the photoelectron energy for an internuclear distance of (a) 2.5 a.u. and (b) 4.0 a.u. For both π- and σ-channels ($\mu=1,0$): $l=0$, dashed curve; $l=1$, single chain curve; $l=2$, solid curve; $l=3$, long dashed curve; $l=4$, triple chain curve.

05 November 2024 20:12:16

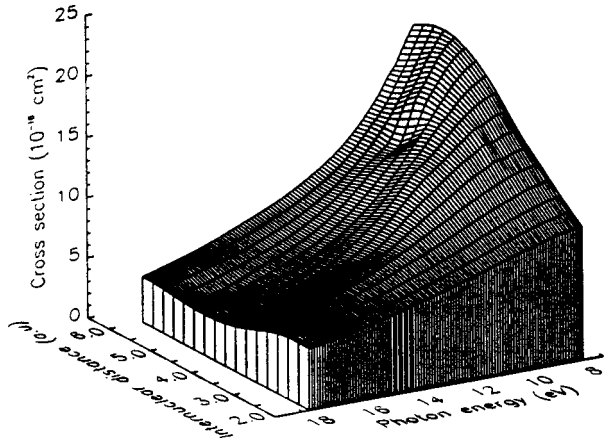


FIG. 4. Photoionization cross section (geometric mean, see text) as function of internuclear distance and photon energy.

In Fig. 5 the vibrationally resolved cross sections $\sigma_{\nu\nu^+}$ for $\nu=0$, 1, and 2 are displayed as function of photon energy, E_{ph} (for $E_{\mathrm{ph}}\gtrsim 18$ eV the energy of the emitted electron might be larger than 10 eV, in which case $\sigma_{\nu\nu^+}$ has been found by extrapolation). From Fig. 5 it is clear that the observed R-dependence of the photoionization cross section does not cause a significant difference with the results obtained in the so-called Franck–Condon approximation. In the Franck–Condon approximation the electronic transition moment is assumed to be independent of the internuclear distance, thus implying that the vibrationally resolved cross sections [see Eq. (9)] are identical but for a constant factor given by the overlap of initial and final vibrational states. This behavior of $\sigma_{\nu\nu^+}$ is evident in Fig. 5. It is due to the limited spatial extension of the LiH($X\,^1\Sigma^+$) vibrational states, that the R-dependence of the transition moment does not influence the vibrationally resolved spectra significantly. For the lowest ten vibrational states, the classical allowed region is within 2 a.u. and 5 a.u. Thus, only for highly excited states is there a chance to observe a non-Franck–Condon behavior of $\sigma_{\nu\nu^+}$, and for experiments under thermal conditions these states will not be populated. The ratios $\sigma_{\nu\nu_1^+}/\sigma_{\nu\nu_2^+}$ that can be extracted from Fig. 5 are furthermore in agreement with the Franck–Condon factors $|\langle\nu|\nu^+\rangle|^2$ calculated in this work, see Table IX. Also listed in Table IX are the Franck–Condon factors for the $\nu^+\leftarrow\nu=0$ transitions previously reported.[10,11] Except for the surprisingly large factor of 0.3413 for the $2\leftarrow 0$ transition quoted by Berriche *et al.*,[10] the three sets of data typically agree within 20%.

In the experimental studies of photoionization of small molecules, the He(I) photoelectron spectroscopy (PES) technique, where the He(I)α line of 21.217 eV is used as a light source, has proven well suited.[40,41] As the structure of LiH($X\,^1\Sigma^+$) is well known, such an experimentally determined energy spectrum of the emitted electron, will contain information about the less known structure of the ground state of LiH^+($X\,^2\Sigma^+$). The analysis of the experimental spectrum has to be guided by theory, and we therefore present the theoretical PE-spectra. In order to compare experimental spectra with theoretical calculations, the population of the rovibrational states given by the experimental

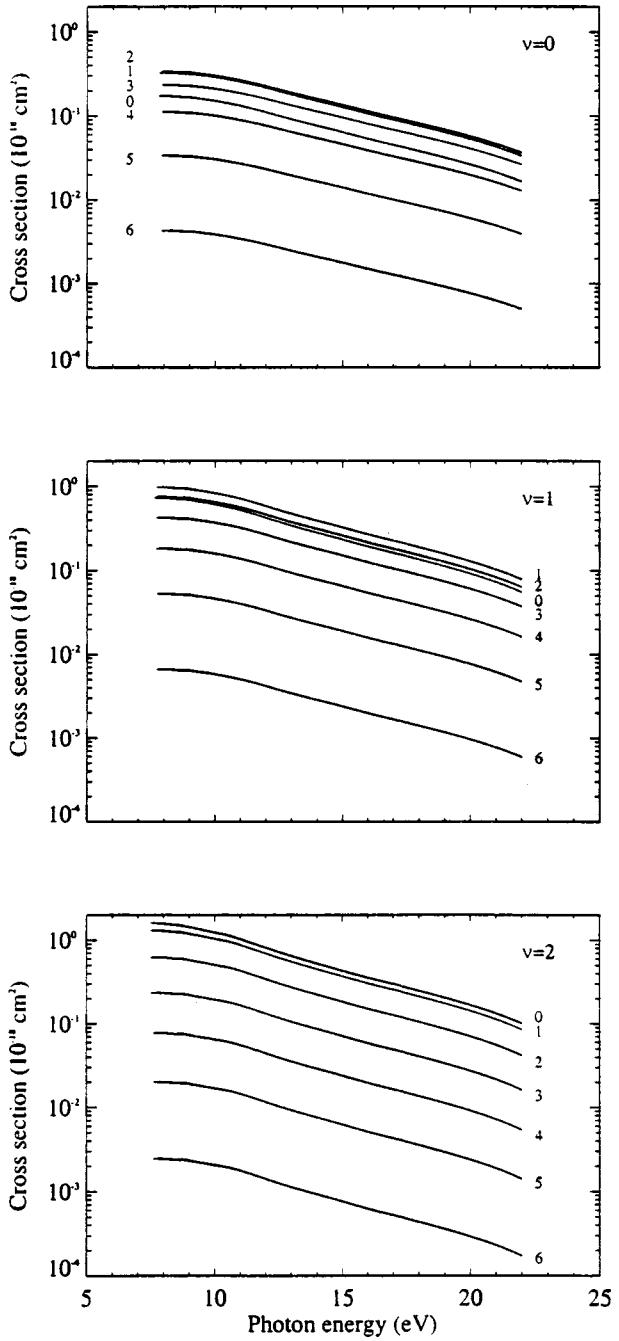


FIG. 5. Vibrationally resolved photoionization cross sections, $\sigma^G_{\nu\nu^+}=\sqrt{\sigma^L_{\nu\nu^+}\sigma^V_{\nu\nu^+}}$ for $N=N^+=0$. Upper panel, $\nu=0$; center panel, $\nu=1$; lower panel, $\nu=2$. The row of numbers listed in each panel indicates the vibrational number ν^+ of the molecular ion.

temperature and the final resolution for the electron detection have to be included in the simulation. In the following it is assumed that the experimental conditions are such that the population of the rovibrational states obeys a Boltzmann distribution. Further, the resolution function of the electron detector has been taken to be Gaussian with a small, but not unrealistic, FWHM value of 10 meV. The simulated cross section, $\tilde{\sigma}(E_{\mathrm{el}},T)$, is given by

$$\tilde{\sigma}(E_{\mathrm{el}},T)=\sum_{\nu N}(2N+1)P_{\nu N}(T)\sigma_{\nu N}(E_{\mathrm{el}}), \tag{10}$$

05 November 2024 20:12:16

TABLE IX. Franck–Condon factors.

ν/ν^+	0	1	2	3	4	5	6
0[a]	0.0121	0.0237	0.0253	0.0158			
0[b]	0.0166	0.0318	0.3413	0.0245	0.0125	0.0037	0.0009
0[c]	0.0153	0.0292	0.0305	0.0214	0.0103	0.0031	0.0004
1[c]	0.0610	0.0823	0.0643	0.0366	0.0157	0.0045	0.0006
2[c]	0.1252	0.1033	0.0497	0.0188	0.0062	0.0016	0.0002
3[c]	0.1749	0.0665	0.0097	0.0003	0.0000	0.0000	0.0000
4[c]	0.1866	0.0146	0.0036	0.0105	0.0068	0.0022	0.0003
5[c]	0.1619	0.0021	0.0349	0.0276	0.0121	0.0035	0.0005

[a]Grabandt *et al.* (Ref. 11).
[b]Berriche and Gadea (Ref. 10).
[c]Present results.

$$\sigma_{\nu N}(E_{\rm el})=\sum_{\nu^+}\sigma_{\nu N,\nu^+ N}(E_{\rm ph},E'_{\rm el})G(E_{\rm el},E'_{\rm el}) + \int_0^\infty \sigma_{\nu N,\varepsilon^+ N}(E_{\rm ph},E'_{\rm el})G(E_{\rm el},E'_{\rm el})dE'_{\rm el}, \qquad (11)$$

where $E'_{\rm el}=E_{\rm ph}+\varepsilon_{\nu N}-\varepsilon_{\nu+N}$ in the summation and $E'_{\rm el}=E_{\rm ph}+\varepsilon_{\nu N}-\varepsilon^+$ in the integration. $P_{\nu N}$, G, and ε^+ denote the Boltzmann factor, the Gaussian distribution function, and the energy of the rovibrational continuum state of LiH^+, respectively. In the calculation we include rovibrational states of $LiH(X\,^1\Sigma^+,\nu N)$ with $N=0-15$ and for each N the six lowest vibrational states ($\nu=0-5$). The states with $N\neq 0$ have been obtained by adding the rotational energy $N(N+1)/2\mu R^2$ term to the relevant potential curves and solving the resulting Schrödinger equation similar to the $N=0$ case.[42] For $N>15$ there are no bound states of $LiH^+(X\,^2\Sigma^+)$, thus all bound–bound transitions are taken into account with the chosen range of N's. Further we note that at $T=1500$ K, the highest temperature considered here, the $(2N+1)$ weighted Boltzmann factors for the $\nu=5$ states are in the order of 10^{-4}, i.e., all states with any appreciable population are included in the calculation. The energies of the continuum states of $LiH^+(X\,^2\Sigma^+)$ cover the range 0.1 meV–2 eV with the grid size gradually varying from 0.05 meV to 25 meV.

The results of the simulations for three typical experimental[12] temperatures of 300 K, 900 K, and 1500 K are shown in Fig. 6 (bottom spectra). At $T=300$ K practically only the ($\nu=0,N$) states of $LiH(X\,^1\Sigma^+)$ are populated and only one group of peaks from bound–bound transitions is seen. Within this group four peaks corresponding to the transitions $\nu=0\rightarrow\nu^+$, $\nu^+=0$, 1, 2, and 3, with the rotational lines $N=1-3$ contributing the most to the peaks, are found at the ionization energies -7.678 eV, -7.722 eV, -7.754 eV -7.776 eV, respectively. As the temperature is increased, contributions from excited vibrational states of $LiH(X\,^1\Sigma^+)$ appear in the spectra, and at a temperature of 1500 K the lines due to transitions from the ($\nu=1,N$) group of states become comparable in strength with $\nu=0$ lines. Summing over the rotational quantum number, the population of the $\nu=0$ and the $\nu=1$ states at this temperature become 0.72 and 0.20, respectively, but the cross sections (as seen in Fig. 5 for the $N=0$ case) for $\nu=1$ are 2–4 times larger than those of $\nu=0$, thus causing the transition strengths to be of the

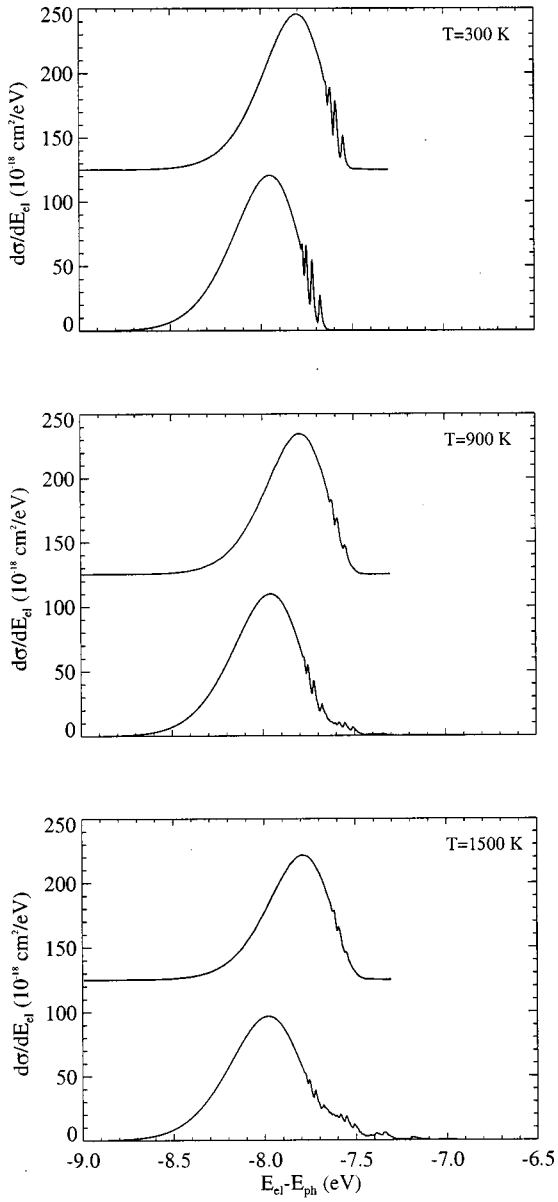


FIG. 6. Simulated photoelectron spectra ($\tilde{\sigma}^G=\sqrt{\tilde{\sigma}^L\tilde{\sigma}^V}$) for the photon energy $E_{\rm ph}=21.217$ eV. Upper panel, $T=300$ K; center panel, $T=900$ K; lower panel, $T=1500$ K. In each panel the lower spectra are based on the present calculations, while the upper are taken from Grabandt *et al.* (Ref. 11). The continuum peaks of the latter have been scaled to the present results and the resulting spectra have been shifted with 125 $\times 10^{-18}$ cm^2/eV.

05 November 2024 20:12:16

same order. That more states contribute to the ionization process at the higher temperatures tends to broaden the peaks in the spectra, thus making it more difficult to extract information on the rovibrational structure of LiH^+. Still, at $T=900\,K$ the present calculations predict some rather distinct peaks, and hence suggest that an experiment measuring the PE-spectrum of LiH will give the wanted information.

Also shown in Fig. 6 are the simulated spectra calculated by Grabandt *et al.*[11] (upper spectra). These calculations differ from the present since they use the Franck–Condon approximation. In order to facilitate the comparison with the present spectra, the continuum peak of the spectra from[11] have been scaled to the height of the continuum peak of the present and shifted with $125\times10^{-18}\,cm^2/eV$. The respective scaling factors are 120 (300 K), 110 (900 K), and 97 (1500 K) in units of $10^{-18}\,cm^2/eV$.

Comparing the present spectra with the spectra of Grabandt *et al.*[11] reveals differences in the position and, to a lesser extent, in the width of the transition lines. The shift in position is mostly due to the difference in the assumed ionization potential of LiH, where the ionization potential is defined as the energy difference at the equilibrium values for the LiH^+ and the LiH potentials curves. The present CI-calculation gives a value of 7.743 eV, whereas a value of 7.612 eV is found from the potential curves given by Grabandt *et al.*[11] These values should be compared with the experimental result of 7.7 eV.[39] Neglecting this energy shift, the present calculation and the calculation of Grabandt *et al.*[11] for $T=300\,K$ result in quite similar spectra. At increasing temperature, however, the agreement between the two sets of calculations becomes less satisfactory. First of all, the transitions from the $\nu=1$ vibrational level of LiH do not appear in the spectra of Grabandt *et al.*[11] Secondly, the peaks due to bound–bound transitions are sharper in the present spectra than in the spectra of Grabandt *et al.*,[11] whereas the opposite holds for the bound-continuum peak. The latter is probably caused by a difference in the description of the ionic potential and the resulting continuum wave functions.

IV. CONCLUSION

Accurate ground state potential curves for LiH and LiH^+ have been calculated. From the LiH potential curve the spectroscopical constants and the vibrational structure have been determined, showing excellent agreement with experimental data. The permanent dipole moment of the ground state of LiH has also been determined, and it is found to vary strongly as a function of internuclear distance due to the influence of the ion-pair curve Li^+-H^-. Further, for the first time, vibrationally resolved photoionization cross sections have been calculated for the LiH molecule. The vibrationally resolved relative cross sections are found to be very similar to the branching ratios determined from the Franck–Condon factors, despite the observed energy and *R*-dependence of the calculated photoionization cross section.

Finally, simulations of experimental photo-electron spectra were presented. Significant deviations were found compared to previous simulations. This is most likely due to differences in the calculated potential curves and the resulting energy differences. Both sets of simulations, however, show that measurements of the photoionized electron may give the first experimental information on the rovibrational structure of the LiH^+ ion.

ACKNOWLEDGMENTS

The authors wish to thank Dr. Paul J. A. Ruttink from the division of theoretical chemistry of the Debye Institute, Utrecht University, for advice and discussions concerning the calculation of the potential curves. M.F.V.L. wishes furthermore to acknowledge support by E. U. (Contract No. ERBFMBICT960592).

[1] G. Nakamura, Z. Phys. **59**, 218 (1930); G. Nakamura and T. Shidei, Jpn. J. Phys. **7**, 33 (1931).
[2] F. H. Crawford and T. Jorgensen, Jr., Phys. Rev. **47**, 358 (1935); **47**, 932 (1935); **48**, 745 (1936).
[3] F. B. Orth and W. C. Stwalley, J. Mol. Spectrosc. **76**, 17 (1979).
[4] Y. C. Chan, D. R. Harding, W. C. Stwalley, and C. R. Vidal, J. Chem. Phys. **85**, 2436 (1986).
[5] W. C. Stwalley and W. T. Zemke, J. Phys. Chem. Ref. Data **22**, 87 (1993).
[6] A. Dalgarno, H. Kirby, and P. C. Stancil, Astron. J. **458**, 397 (1996).
[7] P. C. Stancil, S. Lepp, and A. Dalgarno, Astron. J. **458**, 401 (1996).
[8] H. R. Ihle and C. H. Wu, J. Chem. Phys. **63**, 1605 (1974).
[9] I. Cacelli, R. Moccia, and A. Rizzo, J. Chem. Phys. **102**, 7131 (1995).
[10] H. Berriche and F. X. Gadea, Chem. Phys. **203**, 373 (1996).
[11] O. Grabandt, H. J. Bakker, and C. A. de Lange, Chem. Phys. Lett. **189**, 291 (1992).
[12] T. Ban, H. Skenderović, B. Resan, and G. Pichler, Fiz. A **6**, 161 (1997).
[13] M. F. Guest, P. Fantucci, R. J. Harrison, J. Kendrick, J. H. van Lenthe, K. Scoeffel, and P. Sherwood, GAMESS-UK User's Guide and Reference Manual, CFS Daresbury Laboratory, 1993.
[14] S. Huzinage, J. Chem. Phys. **42**, 1293 (1965).
[15] P. O. Widmark, P. A. Malmqvist, and B. Roos, Theor. Chim. Acta **77**, 291 (1990).
[16] K. R. Lykke, K. K. Murray, and W. C. Lineberger, Phys. Rev. A **43**, 6104 (1993).
[17] *Handbook of Chemistry and Physics*, 76th ed. (1995–1996), edited by D. R. Lide (Chemical Rubber, Cleveland, 1995).
[18] R. S. Mulliken, Phys. Rev. **50**, 1017 (1936).
[19] A. Boutalib and F. X. Gadea, J. Chem. Phys. **97**, 1144 (1992).
[20] H. Partridge and S. R. Langhoff, J. Chem. Phys. **74**, 2361 (1981).
[21] H. Berriche and F. X. Gadea, Chem. Phys. Lett. **247**, 85 (1995).
[22] T. R. Proctor and W. C. Stwalley, J. Chem. Phys. **66**, 2063 (1976).
[23] S. E. Koonin, *Computational Physics* (The Benjamin/Cummings, New York, 1986).
[24] W. C. Stwalley, W. Zemke, K. R. Way, K. C. Li, and T. R. Proctor, J. Chem. Phys. **66**, 5412 (1977).
[25] E. F. Peasrson and W. Gordy, Phys. Rev. **177**, 59 (1969).
[26] K. C. Li and W. C. Stwalley, J. Mol. Spectrosc. **69**, 294 (1978).
[27] C. R. Vidal and W. C. Stwalley, J. Chem. Phys. **77**, 883 (1982).
[28] G. M. Plummer, E. Herbst, and F. C. De Lucia, J. Chem. Phys. **81**, 4893 (1984).
[29] K. K. Docken and J. Hinze, J. Chem. Phys. **57**, 4928 (1972); **57**, 4936 (1972).
[30] W. Meyer and P. Rosmus, J. Chem. Phys. **63**, 2356 (1975).
[31] P. Rosmus and W. Meyer, J. Chem. Phys. **66**, 13 (1977).
[32] M. W. Schmidt, K. K. Baldridge, J. A. Boatz, J. H. Jensen, S. Koseki, M. S. Gordon, K. A. Nguyen, T. L. Windus, and S. T. Elbert, QCPE Bull. **10**, 52 (1990).
[33] R. R. Lucchese, G. Raseev, and V. McKoy, Phys. Rev. A **25**, 2572 (1982).
[34] Aa. Hansen, Mol. Phys. **13**, 425 (1967).
[35] D. K. Watson, T. N. Rescigno, and B. V. McKoy, J. Phys. B **14**, 1875 (1981).
[36] R. R. Lucchese and V. McKoy, Phys. Rev. A **21**, 112 (1980).
[37] H. Rudolph and X. Urbain, Phys. Rev. A **53**, 4111 (1996).
[38] R. R. Lucchese, K. Takatsuka, and V. McKoy, Phys. Rep. **131**, 147 (1985).

05 November 2024 20:12:16

[39] K. P. Huber and G. Herzberg, *Molecular Spectra and Molecular Structure IV. Constants of Diatomic Molecules* (Van Nostrand Reinhold, New York, 1979).
[40] H. J. Bakker, O. Grabandt, and C. A. de Lange, Chem. Phys. **132**, 243 (1989).
[41] G. Jonkers, C. A. de Lange, and J. G. Snijders, Chem. Phys. **50**, 11 (1980).
[42] The measured energies (Ref. 28) 14.809 cm^{-1} and 14.384 cm^{-1} of the rotational transitions $(0,0)\rightarrow(0,1)$ and $(1,0)\rightarrow(1,1)$, respectively, agree well with the corresponding values, 14.815 cm^{-1} and 14.402 cm^{-1}, obtained by the present calculations.

05 November 2024 20:12:16